

第三遍 9/22

Recitation 04

Decision Structure

or → 有True 一定True and → 有False 一定False .

Question 1 | Boolean Expressions

1) Evaluate the following expressions.

False or True and False	False
True and not (9 < 5)	True
24 / 12 >= 2 and 'hello' != 'goodbye'	True
55 != 'hello'	True
'2' == 2	False
'hello' > 35	TypeError

2) Consider $A = \text{True}$ and $B = \text{False}$. What is the resulting value for the following boolean expressions? If there's an error, write error.

(B and A) or A	True
A == (B and (A or (not A)))	False
A == 'True'	False
B == False	True
'True' > B	TypeError

3) Assume that $x = 10$, $y = 20$, and $z = 30$. Evaluate the following boolean expressions. If the expression is always True or False regardless of the values of x , y and z , then add Always to your answer. For example:

- $x == 10$ True
- $x == x$ Always True

$x=10 \ y=20 \ z=30$

(x == 10) and (y > 10)	True
(x < y) and (y > z)	False
(x < 10) or (x > 10)	False
(x < (y + z)) and (x + 10 >= 20)	True
((z - y) == x) and (abs(y - z) == x)	True
x < 10 and x > 10	Always False
x >= y or y >= x	Always True
not (x < (y + z)) or not ((x + 10) <= 20)	False
not (x == y) and (x != y) and (x < y or y < x)	True

T T True .

```
n = input("Enter a Number :")
list = ["I", "II", "III", "IV", "V", "VI", "VII", "VIII", "IX", "X"]
if int(n) > 10:
    print("The number is invalid")
else:
    print("The corresponding Roman numeral is:", list[int(n)-1])
```

Question 2 | Chinese Zodiac

The Chinese Zodiac assigns animals to years in a 12-year cycle. One 12-year is show in the table below. The pattern repeats from there, with 2012 being another year of the dragon, and 1999 being another year of the hare.

Write a program that reads a year from the users and display the animals associated with that year. Your program should work correctly for any year greater than or equal to 1900, not just the ones listed in the table.

Year	Chinese zodiac sign	Year	Chinese zodiac sign
1900	Rat	2000	Dragon
1901	Ox	2001	Snake
...	Tiger	...	Horse
	Hare		Sheep
	Dragon		Monkey
	Snake		Rooster
	Horse		Dog
	Sheep		Pig
	Monkey		Rat
	Rooster		Ox
	Dog		Tiger
1999	Pig	2011	Hare

```
Year=input("Enter a int(Year): ")
if int(Year) < 1900:
    print("int(Year) must be greater than or equal to 1900")
else:
    if (int(Year)-1900) % 12 == 0:
        print("int(Year) of the Rat")
    elif (int(Year)-1900) % 12 == 1:
        print("int(Year) of the Ox")
    elif (int(Year)-1900) % 12 == 2:
        print("int(Year) of the Tiger")
    elif (int(Year)-1900) % 12 == 3:
        print("int(Year) of the Rabbit")
    elif (int(Year)-1900) % 12 == 4:
        print("int(Year) of the Dragon")
    elif (int(Year)-1900) % 12 == 5:
        print("int(Year) of the Snake")
    elif (int(Year)-1900) % 12 == 6:
        print("int(Year) of the Horse")
    elif (int(Year)-1900) % 12 == 7:
        print("int(Year) of the Goat")
    elif (int(Year)-1900) % 12 == 8:
        print("int(Year) of the Monkey")
    elif (int(Year)-1900) % 12 == 9:
        print("int(Year) of the Rooster")
    elif (int(Year)-1900) % 12 == 10:
        print("int(Year) of the Dog")
    elif (int(Year)-1900) % 12 == 11:
        print("int(Year) of the Pig")
    else:
        print("Invalid int(Year) input")
```

Question 3 | Roman Numerals

Write a program that prompts the user to enter a number within the range of 1 through 10.\ The program should display the Roman numeral version of that number. If the number is outside the range of 1 through 10, the program should display an error message.

The following table shows the Roman numerals for the numbers 1 through 10:

Roman numeral	I	II	III	IV	V	VI	VII	VIII	IX	X
Value	1	2	3	4	5	6	7	8	9	10

Question 4 | Roulette Wheel

The pockets of the roulette wheel are numbered from 0 to 36.

- There is a green pocket numbered 0 (zero).
- In number ranges from 1 to 10 and 19 to 28:
 - odd numbers are red and even are black.
- In ranges from 11 to 18 and 29 to 36,
 - odd numbers are black and even are red.



- 1) Write a program that asks the user to enter a pocket number and displays whether the pocket is green, red, or black. The program should display an error message if the user enters a number that is outside the range of 0 through 36.

Question 5 | Time calculator

Write a program that asks the user to enter a number of seconds in years, days, minutes, and seconds left.

- There are 60 seconds in a minute.
- There are 3,600 seconds in an hour.
- There are 86,400 seconds in a day.
- There are 31,536,000 seconds in a year.

```
n = input("Enter a number :")
if int(n) > 36 or int(n) <= 0:
    print("The number is invalid")
else:
    if 1 <= int(n) <= 10 or 19 <= int(n) <= 28:
        if int(n) % 2 == 0:
            print("the color is Black")
        else:
            print("the color is Red")
    elif 11 <= int(n) <= 18 or 29 <= int(n) <= 36:
        if int(n) % 2 == 0:
            print("the color is Red")
        else:
            print("the color is Black")
```

```
n = input("Enter a Number :")
y = int(n)//31536000
d = (int(n)%31536000)//86400
h = ((int(n)%31536000)%86400)//3600
m = (((int(n)%31536000)%86400)%3600)//60
s = (((int(n)%31536000)%86400)%3600)%60
print(y, "years", d, "days", h, "hours", m, "minutes", s, "seconds")
```